

1.1 HEALTH AND HEAT

1.1.1 EXPOSURE OF VULNERABLE POPULATIONS TO HEATWAVES

Exposure to extreme heat has a range of health consequences including, heat stress and heat stroke, worsening heart disease, and acute kidney injury. Infants and older adults are particularly vulnerable to adverse health effects from heat exposure and are being increasingly exposed to heatwaves, defined as a period of 2 or more days where both the minimum and maximum temperatures are above the 95th percentile of 1986–2005. This indicator tracks the number of heatwave days and the exposure of these vulnerable populations, those under 1 and over 65, to heatwaves.

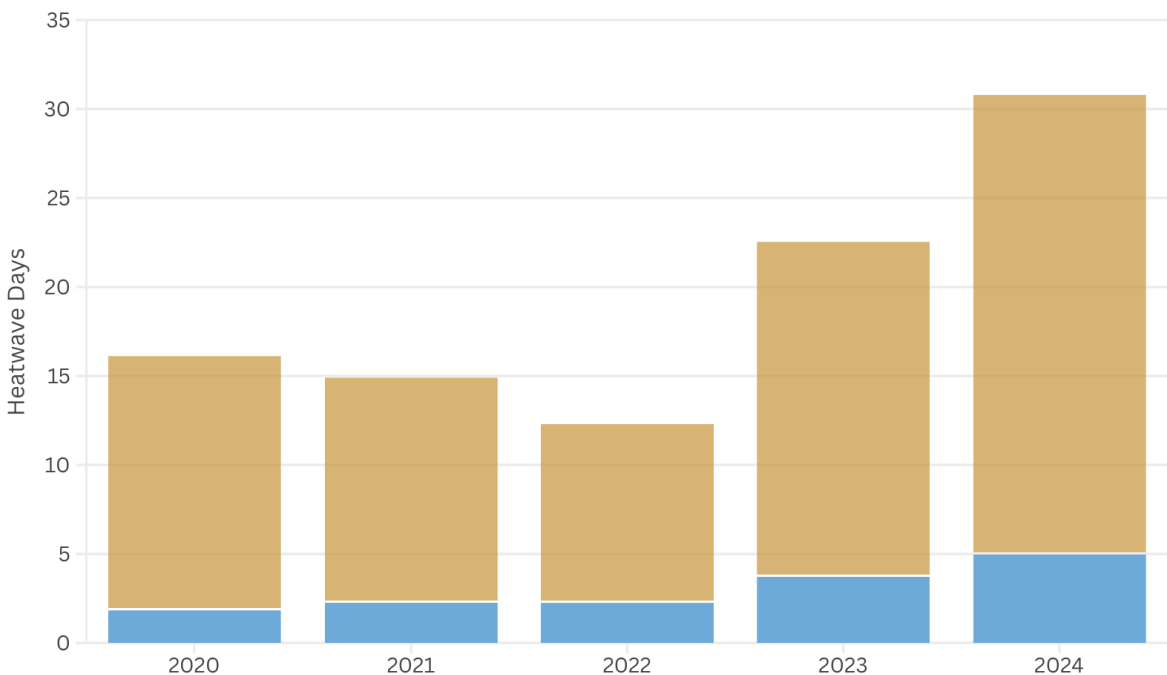
Heatwave Days Attributable to Climate Change

Number of heatwave days attributable to human-induced climate change between 2020 and 2024

The sum of expected exposure days and days attributable to climate change gives the total average number of heatwave days that people were exposed to in a given year.

Global

- Expected exposure to heatwave days without climate change
- Exposure to heatwave days attributable to climate change



Please reference the 2025 Report of the Lancet Countdown if using this data • For a full description of the indicator, see the 2025 report of the Lancet Countdown at lancetcountdown.org



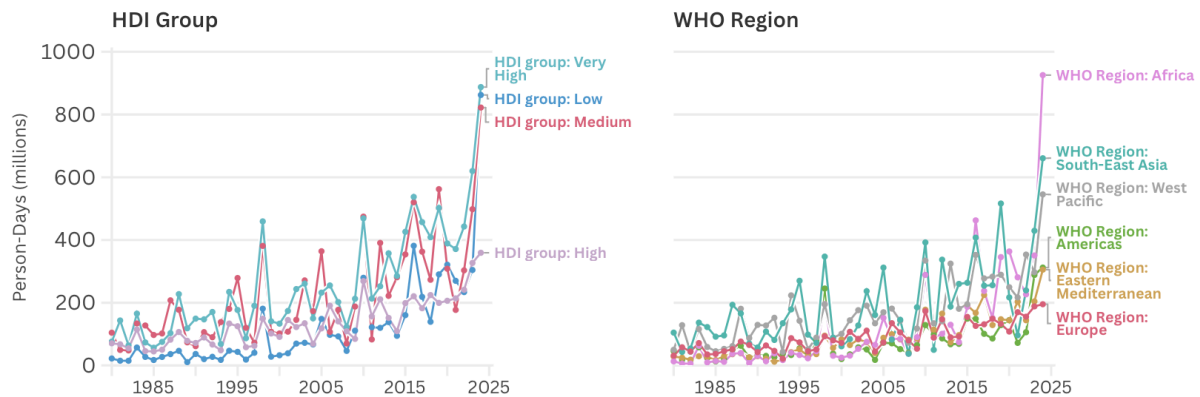
Exposure of Vulnerable Populations to Heatwaves

Number of heatwave person-days experienced by vulnerable groups, by Human Development Index (HDI) group and World Health Organization (WHO) region

Person-days refer to the cumulative number of days of heatwave that people were collectively exposed to (e.g., if 100 people are each exposed for 5 days, there would be 500 person-days)

Infants **Over 65 years old**

Enter HDI group or WHO region to show



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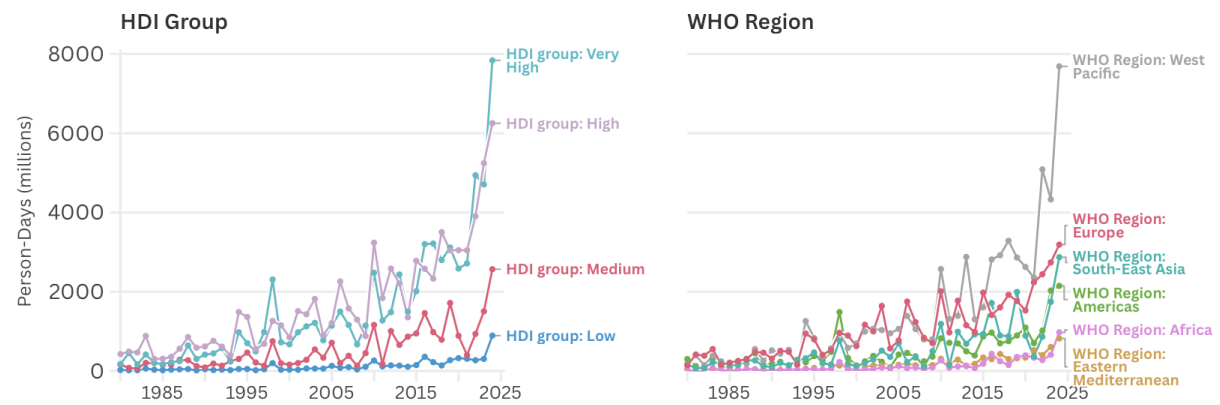
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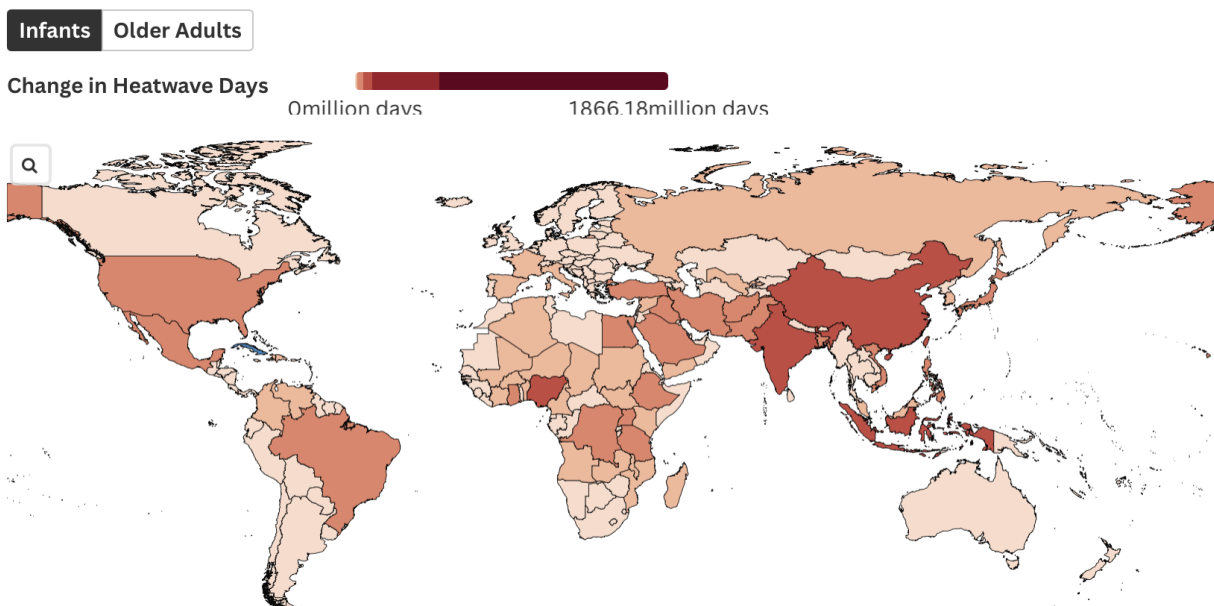


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Exposure of Vulnerable Populations to Heatwaves

Change in average annual number of heatwave person-days experienced by infants (under 1 year old) and older adults (>65 years), comparing 1986-2005 and 2015-2024

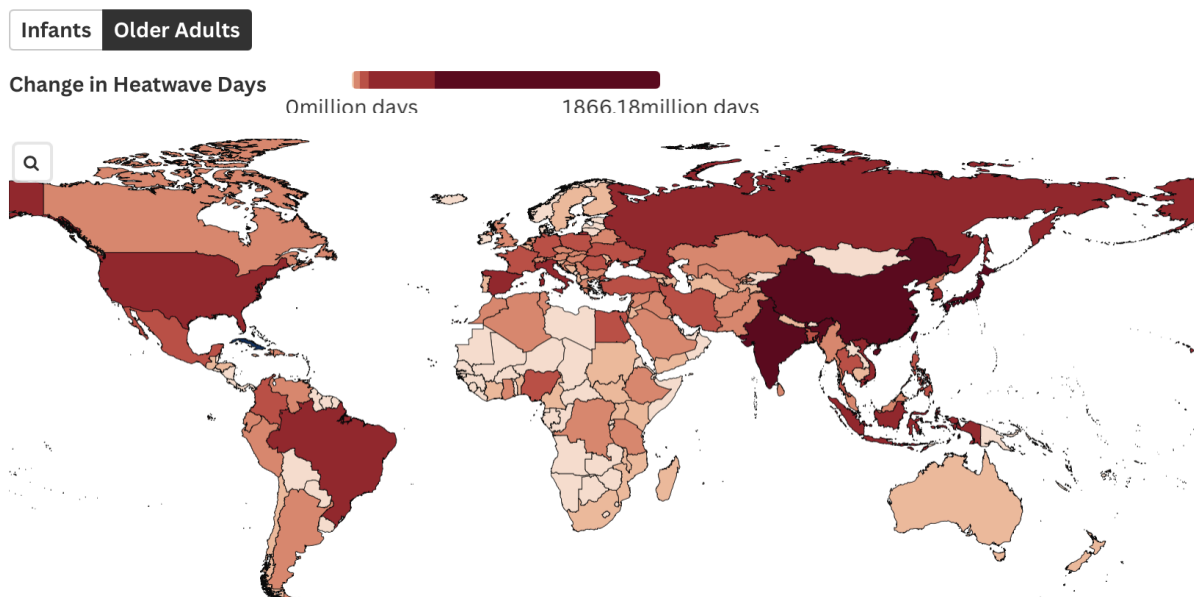


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HEADLINE FINDINGS

In 2024, people older than 65 years and infants younger than 1-year experienced, on average, 304% and 389% more days of heatwaves per person, respectively, compared to 1986-2005.

CAVEATS

As two distinct sources were used for population data to obtain estimates of both the spatial and temporal characteristics there may be some inconsistencies between the pre and post 2000 values.

This indicator was last updated in October 2025

INDICATOR AUTHORS

- 1) Dr. Andrew J Pershing, Joseph Giguere
- 2) Dr Federico Tartarini, Prof Ollie Jay

DATA SOURCES

1. European Centre for Medium-Range Weather Forecasts Reanalysis 5th Generation monthly averaged data on single levels from 1959 to present. Copernicus Climate Change Service Climate Data Store.
2. Hybrid gridded demographic data for the world (1980-2000) 0.25° resolution, Lancet Countdown 2023
3. WorldPop Age and Sex Structure Unconstrained Global Mosaic data for 2000–2020

INDICATOR DESCRIPTION

This indicator tracks changes in the number of heatwave days per year per vulnerable person and reflects the exposure of vulnerable populations to heatwaves expressed as person-days, this captures increasing durations of heatwaves as well as changes in frequency compared with the average number of events in the reference period (1986–2005).

DATA DOWNLOAD

[Click here to download the data on heatwave days attributable to climate change.](#)

[Click here to download the data on vulnerable populations exposed to heatwave days.](#)

1.1.2 HEAT AND PHYSICAL ACTIVITY

Physical activity is key to good health and wellbeing. Hot weather reduces the likelihood of engaging in exercise and increases heat illness risk when it is undertaken. This indicator tracks the impact of a warming world on the number of hours of physical activity potentially lost due to heat.

HEADLINE FINDINGS

In 2024, each person was exposed, on average, to a record-high 1,609 hours during which ambient heat posed at least a moderate heat stress risk during light outdoor exercise, 35.8% above 1990-1999.

CAVEATS

The estimation of heat stress risk for a given exercise category may not be uniform across the entire population, and risk estimates in particular may be different for older adults, young children, pregnant women, and those living with disabilities or chronic diseases. A more detailed interpretation model of heat effects on exercise would incorporate individual factors such as age, health status, physiology, and clothing.

This indicator was last updated in October 2025

INDICATOR AUTHORS

Dr Troy J Cross, Dr Samuel H Gunther, Prof Ollie Jay, Dr Jason KW Lee

DATA SOURCES

1. European Centre for Medium-Range Weather Forecasts Reanalysis 5th Generation monthly averaged data on single levels from 1959 to present. Copernicus Climate Change Service Climate Data Store.
2. Gridded Population of the World Version 4, 2021. Socioeconomic Data and Applications Center, National Aeronautics and Space Administration.
3. World Health Organisation Detailed Boundary ADM0 Shapefiles

INDICATOR DESCRIPTION

This indicator incorporates temperature, humidity, and solar radiation restricting analysis to local sunlight hours only to estimate and track the daily hours per person during which undertaking outdoor physical activity would pose a heightened heat stress risk. The indicator uses “moderate” heat stress risk, as defined by the 2021 Sports Medicine Australia Extreme Heat Policy, which stratifies estimated heat stress risk based on ambient temperature and relative humidity.

DATA DOWNLOAD

[Click here to download indicator data.](#)

1.1.3 CHANGE IN LABOUR CAPACITY

Our capacity to work is affected by temperature and humidity, particularly in highly physical jobs in poorly cooled environments or outdoors with little to no shade in agriculture, industry, and manufacturing sectors. Reduced work productivity puts a strain on the socioeconomic determinants of health for individuals and communities. As the world continues to warm, this indicator tracks the change in potential work hours lost due to exposure to high temperatures.

HEADLINE FINDINGS

Heat exposure led to the loss of 640 billion potential labour hours in 2024, 98% above the 1990–1999 annual average

DATA SOURCES

1. European Centre for Medium-Range Weather Forecasts Reanalysis 5th Generation monthly averaged data on single levels from 1959 to present. Copernicus Climate Change Service Climate Data Store.
2. Gridded Population of the World (2018). Socioeconomic Data and Applications Center, National Aeronautics and Space Administration.
3. International Labour Organization International Statistics Database (ILOSTAT). ILO. Accessed in 2025
4. The Inter-Sectoral Impact Model Intercomparison Project. ISIMP3b Bias Adjustment. 2022.
5. Global, regional, national, and subnational occupational exposure to solar ultraviolet radiation for 195 countries/areas, and the global, regional, and national attributable burden of non-melanoma skin cancer for 183 countries, 2000-2019: a systematic analysis from WHO/ILO Joint Estimates of the Work-related Burden of Disease and Injury. 2023. Pega F, Momen NC, Stricher KN et al.
6. UN Population Division, World Population Prospects 2024

CAVEATS

The distribution of agricultural, construction manufacturing, and service sector workers used are country averages, applied evenly to the population of each grid cell, thus not accounting for sub-national variation in industry sector.

The numbers and percentages of workers captures those in the formal economy for regions and countries, unpaid work, to which women often dedicate more time than men, is not accounted for.

This indicator was last updated in October 2025

INDICATOR DESCRIPTION

This indicator has two distinct parts. The first monitors the number of outdoor workers (a population at risk) with estimates produced by WHO staff. The second calculates hours of work lost as a result of heat exposure in four sectors: agriculture, construction, manufacturing, and service. It monitors potential hours lost by linking Wet Bulb Globe Temperature (including temperature, humidity and solar radiation) with the amount of energy expended by workers through the typical metabolic rate of workers. It then combines this calculation with the proportion of people working over 15 years old in each of the four sectors in each country to estimate the potential work hours lost per year.

DATA DOWNLOAD

[Click here to download data on potential work hours lost.](#)

[Click here to download data on outdoor workers.](#)

INDICATOR AUTHORS

- 1) Dr Natalie C. Momen, Dr Frank Pega
- 2) Chris Freyberg, Dr Bruno Lemke, Matthias Otto

1.1.4 RISING NIGHTTIME TEMPERATURES AND SLEEP LOSS

Sleep of adequate duration and quality is important for good human physical and mental health. High ambient temperatures are associated with worse sleep quantity and quality. With climate change resulting in nighttime temperatures rising faster than daytime temperatures in many world regions, the risk of adverse health outcomes from poor sleep quality is rising globally.

This indicator tracks the impact of suboptimal nighttime temperatures on sleep loss. It combines the global functional sleep response to nighttime temperature identified in a multi-country sleep study, with nighttime temperature data, and it statistically controls for individual-level demographic and environmental factors, including air conditioning access.

HEADLINE FINDINGS

Sleep hours lost due to high temperatures increased by 6% between 1986-2005 and 2020-2024, reaching a record 9% increase in sleep hours lost in 2024

DATA SOURCES

1. European Centre for Medium-Range Weather Forecasts Reanalysis 5th Generation monthly averaged data on single levels from 1959 to present. Copernicus Climate Change Service Climate Data Store.
2. Gridded Population of the World Version 4, 2021. Socioeconomic Data and Applications Center, National Aeronautics and Space Administration.
3. Rising temperatures erode human sleep globally. One Earth. 2022. Minor K, Bjerre-Nielsen A, Jonasdottir SS, Lehmann S, Obradovich N.

CAVEATS

As with other statistical samples, the indicator's analytic approach assumes that the estimates obtained in the underlying global study of tens of thousands of individuals are likely to apply to the individuals not within the sample. This means that the original data may be more likely to include wealthier persons who can afford fitness trackers and less likely to include the world's poorest than a truly representative sample, potentially underestimating the impact. Additionally, the relationship between ambient temperatures and human sleep may change in the

future as individuals adapt and/or decompensate to altered future temperatures. An increased availability of air conditioning in the future, for example, might alter the global response curve. It is also possible, given the spatial resolution of the indicator, that areas with high air conditioning/fan penetration already reflect some expected adaptation particularly at the higher end of the temperature distribution.

This indicator was last updated in October 2025

INDICATOR DESCRIPTION

This indicator tracks the relationship between human sleep and warmer nighttime temperatures employing estimates from published research that monitors fitness tracking bands to measure over ten billion sleep observations across over 40,000 individuals living in 68 countries around the world between 2015–2017. The indicator estimates percentage change in annual total number of hours of sleep lost globally due to warmer than optimal nighttime temperature relative to the 1986-2005 baseline average in this metric.

DATA DOWNLOAD

[Click here to download indicator data.](#)

INDICATOR AUTHORS

Dr Kelton Minor and Dr Nick Obradovich

1.1.5 HEAT-RELATED MORTALITY

Populations are finding themselves increasingly vulnerable to extreme heat. Exposure to extremes of heat can result in a range of health consequences, including heat stress and heat stroke, worsening heart disease, and acute kidney injury and leads to an increase in all-cause mortality. As the world continues to warm and populations age, this indicator tracks heat-related mortality around the world and the proportion of those that were made more likely by human-induced climate change.

HEADLINE FINDINGS

In 2012-2021, global heat-related mortality reached an estimated average 546,000 deaths annually, up 63.2% from 335,000 in 1990-1999.

DATA SOURCES

1. European Centre for Medium-Range Weather Forecasts Reanalysis 5th Generation monthly averaged data on single levels from 1959 to present. Copernicus Climate Change Service Climate Data Store.

2. Gridded Population of the World Version 4, 2021. Socioeconomic Data and Applications Center, National Aeronautics and Space Administration.
3. Yearly, country-level mortality and population data obtained from the Global Burden of Disease 2021
4. Mortality data from United Nations Population Division EUROSTAT
5. All-cause mortality data from the World Mortality dataset, 2015
6. The World Bank Group. World Development Indicators. 2015

CAVEATS

The analysis for heat-related mortality assumes the exposure-response function is constant. It does not capture changes in response to heat exposure that might happen over time, as a result of acclimation and adaptation. Not capturing these changes could result in an over-estimation of heat-related deaths in later calendar years. Annual average mortality rates are used, rather than daily mortality rates. Given baseline mortality can be higher in colder months, this may lead to an overestimation of overall mortalities.

This indicator was last updated in October 2025

INDICATOR DESCRIPTION

This indicator monitors heat-related mortality using a newly-developed model framework that builds on a mortality database for 120 countries. It then applies state-of-the-art meta-prediction models to consistently estimate the association between temperature and mortality from all causes in all countries globally, and combines them with yearly mortality estimates from the Global Burden of Disease, making it the most comprehensive global estimate of heat-related mortality yet.

DATA DOWNLOAD

[Click here to download indicator data.](#)

INDICATOR AUTHORS

Dr Joan Ballester, Prof Xavier Basagaña, Jorge Ruiz-Cabrejos

1.2.5 Extreme Weather and Sentiment

1.2.5 EXTREME WEATHER AND SENTIMENT

Climate change-related increases in extreme weather events, including heatwaves and extreme rainfall, pose diverse risks to mental health globally, ranging from altered affective states to elevated mental health-related hospitalisations and suicidality. As the world continues to warm, this indicator

tracks people's sentiment expressed on social media in relation to extreme weather events.

Health-related data fields that could be used in mock cases, with a data-minimization mindset.

<https://censinet.com/perspectives/how-to-implement-phi-data-minimization-in-healthcare>